

Programming Languages
(Course code : ACS101)

Lecture notes on Compilation, Linking and Execution of C Program

Compiler and Interpreter

High-level Programming language is not directly understood by the Computer. It has to be translated into Machine language (Machine code) to be understood by the Computer for execution.

Compiler is a program (software) that converts an entire source code (program) written in a High-level language into machine readable code as executable file or Object code for the target machine (Computer).

Interpreter is a software that translates a source program (code) line-by-line (step-by-step) to machine –readable instructions for execution by Computer. An Interpreter translates source code line-by-line (step-by-step) on the fly, and does not generate executable file.

Compilation process of C Program

The Compilation process for a C program is the sequence of steps that transforms human-readable C source code into machine-executable binary code (0s and 1s).

The entire Compilation process consists of four distinct stages, namely **Pre-processing, Compilation, Assembly, and Linking**, as shown in the **Figure - 1** on the next page.

Loading and Execution of Program

Loading. Operating System Loader loads the executable file (in Machine code) in the Main memory of the Computer for the next stage of Execution.

Execution. The Central Processing Unit (CPU) fetches instructions from the Main memory of the Computer and executes them one-by-one.

The steps for Pre-Processing, Compilation, Assembler, Linking, Loading and Execution of a C Program is shown in **Figure-2** on the next page.

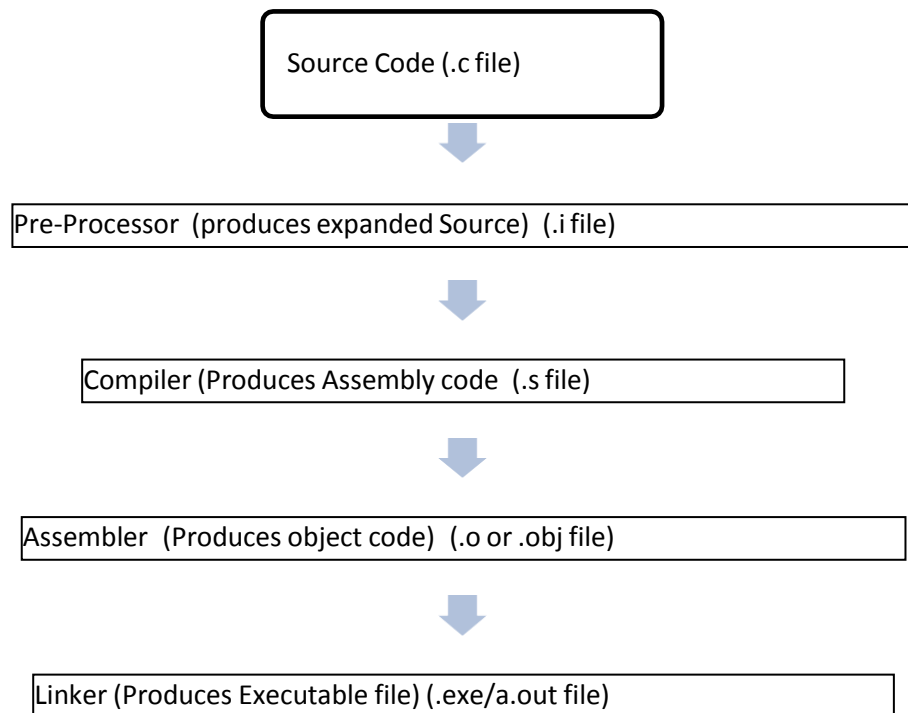


Figure -1 : Stages of Compilation process of C program

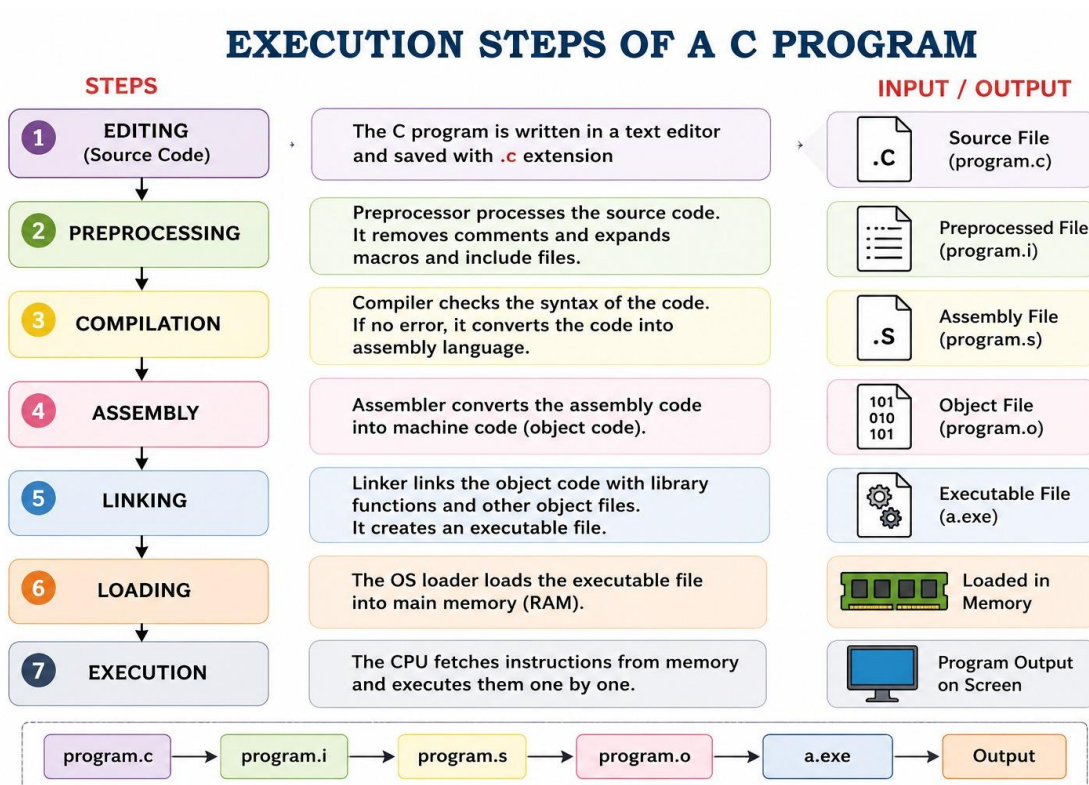


Figure -2 : Steps for Pre-Processing, Compilation, Assembly, Linking, Loading and Execution of a C Program

